

Definition: The Unit Step Function

The unit step function $u(t)$ is defined as

$$u(t) = \begin{cases} 0, & \text{when } t < 0 \\ 1, & \text{when } t > 0. \end{cases}$$

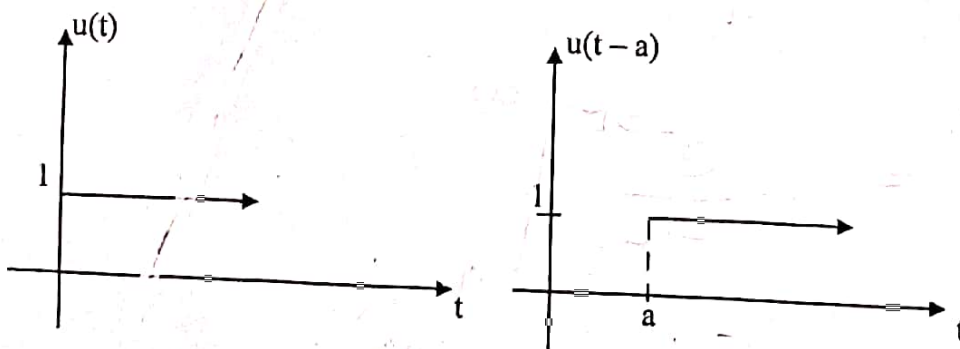
By shifting the argument of $u(t)$ the jump can be moved to a different location. Accordingly we define the Translated step Function as under:

Translated Step Function.

It is defined as,

$$u(t-a) = \begin{cases} 0, & \text{when } t-a < 0 \\ 1, & \text{when } t-a > 0 \end{cases} = \begin{cases} 0, & \text{when } t < a \\ 1, & \text{when } t > a \end{cases} \quad (8.13.1)$$

It has a jump at $t = a$. Often the unit step functions are simply called the *step functions*. They are shown in Fig 8.13.1.



(a) unit step function

(b) Translated Step Function

Fig 8.13.1

Unit Step Function

Ex $L[u(t-a)] = ??$

Sol $L[u(t-a)] = \int_0^{\infty} e^{-st} u(t-a) dt \quad (1)$

$$= \int_0^a e^{-st} u(t-a) dt + \int_a^{\infty} e^{-st} u(t-a) dt \quad (2)$$

$$u(t-a) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t > a \end{cases} \quad (3)$$

$$= \int_a^{\infty} e^{-st} (1) dt \quad (4)$$

$$= \left. \frac{e^{-st}}{-s} \right|_a^{\infty} \quad (5)$$

$$= -\frac{1}{s} (e^{-\infty} - e^{-as}) \quad (6)$$

$$\boxed{L[u(t-a)] = \frac{e^{-as}}{s}} \quad (7)$$

Theorem

Second shift theorem or t shifting

$$\text{Let } \boxed{L[f(t)] = F(s)} \quad \text{--- (1)}$$

$$\text{then, } L[f(t-a)u(t-a)] = e^{-as} L[f(t)] \quad \text{L(2)}$$

$$\Rightarrow \boxed{L[f(t-a)u(t-a)] = e^{-as} F(s)} \quad \text{L(3)}$$

then, Laplace inverse is,

$$\boxed{f(t-a)u(t-a) = L^{-1}[e^{-as} F(s)]}$$

$$\text{L(4)}$$

Ex $L^{-1} \left[\frac{1 - e^{-3s}}{s^2} \right] = ?$

L04

Sol $F(s) = \frac{1 - e^{-3s}}{s^2} = \frac{1}{s^2} - \frac{e^{-3s}}{s^2} \quad (2)$

$\Rightarrow L^{-1}[F(s)] = L^{-1}\left[\frac{1}{s^2}\right] - L^{-1}\left[e^{-3s} \frac{1}{s^2}\right] \quad (3)$

As, $L[t^n] = \frac{n!}{s^{n+1}} \Rightarrow t^n = L^{-1}\left[\frac{1}{s^{n+1}}\right] \quad (4)$

$\Rightarrow f(t-a)u(t-a) = L^{-1}[e^{-as}F(s)] \quad (5)$

$a=3, F(s) = \frac{1}{s^2}$

$L^{-1}\left[\frac{1}{s^2}\right] = t$

$f(t-3) = t-3$
 L^{-1}

$\Rightarrow f(t) = t - (t-3)u(t-3)$

L^{-1}

$u(t-3) = \begin{cases} 0 & \text{if } t < 3 \\ 1 & \text{if } t > 3 \end{cases}$

Ex #02

L5

Find inverse Laplace transform of

$$F(s) = \frac{2}{s^2} - \frac{2}{s^2} e^{-2s} - 4 \frac{e^{-s}}{s} + \frac{s e^{-2\pi s}}{s^2 + 1}$$

L(1)

Sol As ---

$$L[t] = \frac{1}{s^2}, \quad L[1] = \frac{1}{s}, \quad L[\cos t] = \frac{s}{s^2 + 1}$$

$$L^{-1}[e^{as} F(s)] = f(t-a) u(t-a) \quad (2) \quad u(t-a) = L^{-1}\left[\frac{e^{as}}{s}\right]$$

$$\Rightarrow L^{-1}[F(s)] = 2L^{-1}\left[\frac{1}{s^2}\right] - 2L^{-1}\left[\frac{e^{-2s}}{s^2}\right] - 4L^{-1}\left[\frac{e^{-s}}{s}\right] + L^{-1}\left[e^{-2\pi s} \cdot \frac{s}{s^2 + 1}\right] \quad (3)$$

$$f(t) = 2t - 2(t-2)u(t-2) - 4u(t-2) + \cos(t-2\pi)u(t-2\pi)$$

L(4)

Exercice

$$\textcircled{1} \quad \mathcal{L}^{-1} \left[\frac{e^{-4s}}{s+4} \right]$$

$$\textcircled{2} \quad \mathcal{L}^{-1} \left[\frac{s + e^{-\pi s}}{s^2 + 1} \right]$$

$$\textcircled{3} \quad \mathcal{L}^{-1} \left[\frac{e^{-s} - 2e^{-2s} + 2e^{3s} - e^{4s}}{s} \right]$$